Nikhil Narayana

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EDUCATION

Michigan State University, College of Engineering Graduated May 2026
Bachelor of Science, Mechanical Engineering GPA: 3.4
Minor in Aerospace

SKILLS & INTERESTS

Engineerng Software: FEA ANSYS, Siemens NX, MATLAB, SolidWorks, Abaqus, Python, JavaScript
Interests: Energy efficiency in Automotive Design, Aerospace Systems, Emerging technologies, Experimental work
Soft Skills: Time Management, Effective Communication, Problem Solver, Leadership Under Pressure, Multilingual

EXPERIENCE

Michigan State University May 2024 – August 2024
Engineering Intern

- Designed and constructed a **prototype rocket**; conducted 5+ test launches and implemented full-body design improvements, increasing **overall performance and flight stability by 25%**
- Led workshops focused on improving communication, teamwork, and problem-solving throughout the project process skills in addition to **engineering fundamentals**
- Researched **aerodynamic performance** and system design to optimize launch readiness procedures by 25%

Formula SAE – Fuel Systems May 2023 – December 2023
Fuel System Lead

- **Led a 15-member team** to design and integrate a high performance fuel system for Formula SAE vehicles
- Increased safety and efficiency by **reducing fuel leaks by 20%** through insulated fuel lines and optimized regulators
- Collaborated across subsystem teams , improving overall vehicle reliability and cutting integration for successful results

Automotive Technology Department August 2021 – December 2023
Teaching Assistant

- **Mentored and guided a class of 35+ students**, strengthening technical understanding and improving lab performance
- Optimized use of tools, materials, and equipment to reduce downtime and maintain 100% readiness for lab sessions
- Diagnosed and resolved **50+ mechanical issues** on training vehicles, improving system reliability and overall learning of mechanical systems

ACADEMIC PROJECTS

Medical Training Cart Redesign August 2025 – December 2025

- Redesigned a medical training cart for **MSU IMPART Alliance**, improving equipment protection, maneuverability, and transport fit
- Built a **SolidWorks** assembly using aluminum T-slot extrusions, cross-bracing, caster wheels, and panel mounts
- Validated the frame through **ANSYS FEA**, showing only **~25 μm deformation** under typical pushing loads

CrankSlider Project May 2025 – June 2025

- **Led a 4-member team** to design and build a mechanism simulating piston-crankshaft dynamics
- Applied CAD modeling and prototyping to improve efficiency, achieving **15% smoother motion**
- Verified performance using mechanical analysis, improving accuracy and functionality

Slingshot Project July 2025 – August 2025

- **Engineered and fabricated** a 4-bar kicking mechanism using aluminum; completed full prototype from CAD
- Performed **finite element analysis (FEA)** and calculations, validating strength and power efficiency
- Achieved a **ball launch distance of 432 ft**, demonstrating the effectiveness of adjustable power and precise geometry

ACTIVITIES AND LEADERSHIPS

MSU Rocketry | *Manufacturing and Design Team Member* August 2024 – April 2026

- Applied **CAD tools(Siemens NX)** to support design and manufacturing efforts across a **100+ member team**
- Contributed to aerospace development by creating and testing drone prototypes for research and design initiatives
- Produced promotional materials that enhanced outreach and boosted recruitment within the organization